The Cretan human pangenome



Panos Ioannidis, PhD

HSCBB26 conference
Kalamata
23-25-Oct-2026

Collaborators

- Biodiversity computing group
  - Alexis Stamatakis
- Ancient DNA lab
  - Nikos Poulakakis
  - Pavlos Pavlidis
- Computations
- Funding

FORTH
INSTITUTE OF COMPUTER SCIENCE







ANCIENT DNA LAB
FORTH - IMBB
PALEOGENOMICS AND EVOLUTIONARY GENETICS GROUP

HITS
Heidelberg Institute for Theoretical Studies

European Commission

H.F.R.I.
Hellenic Foundation for Research & Innovation

Greece 2.0
NATIONAL RECOVERY AND RESILIENCE PLAN

Funded by the European Union
NextGenerationEU

Grant Agreement: 101087081 (Comp-Bio-div-GR) **Project Number:** 14779 (THESEUS)

Human genomics

- Many human genome sequencing projects
  - *Denmark* (Maretty et al 2017), *Iceland* (Beyter et al 2021), *China* (Li et al 2021), *Africa* (Sherman et al 2019), *Arabs* (Nassir et al 2025), *Pacific* (Littlefield et al 2024)
- Study **genomic** diversity in human populations
- **Facilitated** by the decreasing sequencing **cost**

Human pangenome

- Liao et al 2023: 94 haplotypes

Article | Open access | Published: 10 May 2023

A draft human pangenome reference

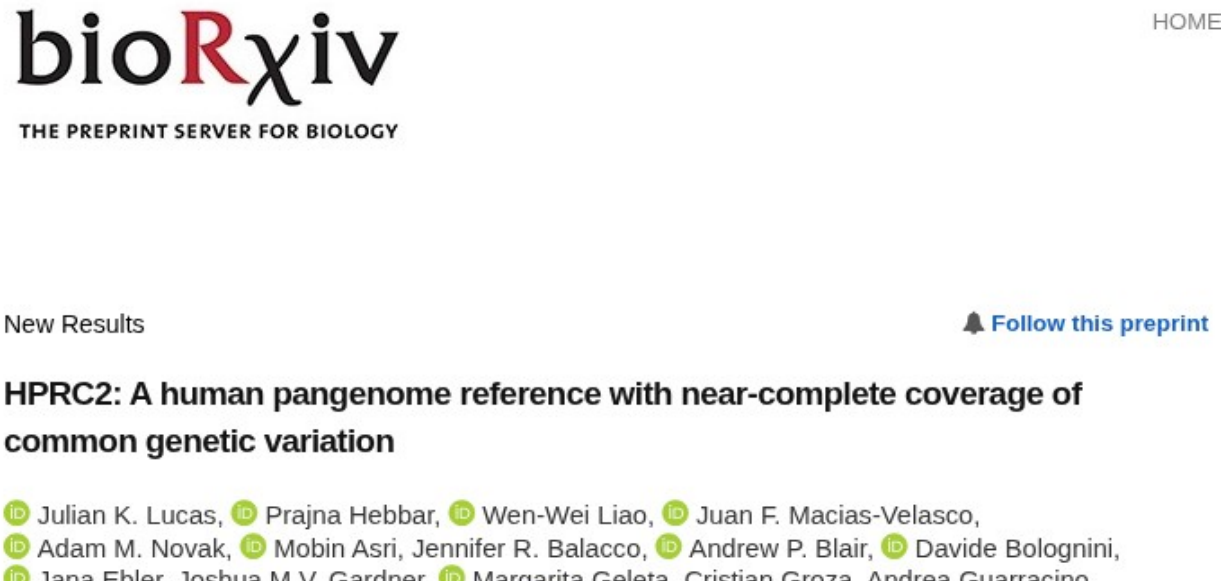
Wen-Wei Liao, Mobin Asri, Jana Ebler, Daniel Doerr, Marina Haukness, Glenn Hickey, Shuangjia Lu, Julian K. Lucas, Jean Monlong, Haley J. Abel, Silvia Buonaiuto, Xian H. Chang, Haoyu Cheng, Justin Chu, Vincenza Colonna, Jordan M. Eizenga, Xiaowen Feng, Christian Fischer, Robert S. Fulton, Shilpa Garg, Cristian Groza, Andrea Guarracino, William T. Harvey, Simon Heumos, … Benedict Paten

+ Show authors

Nature **617**, 312–324 (2023) | Cite this article

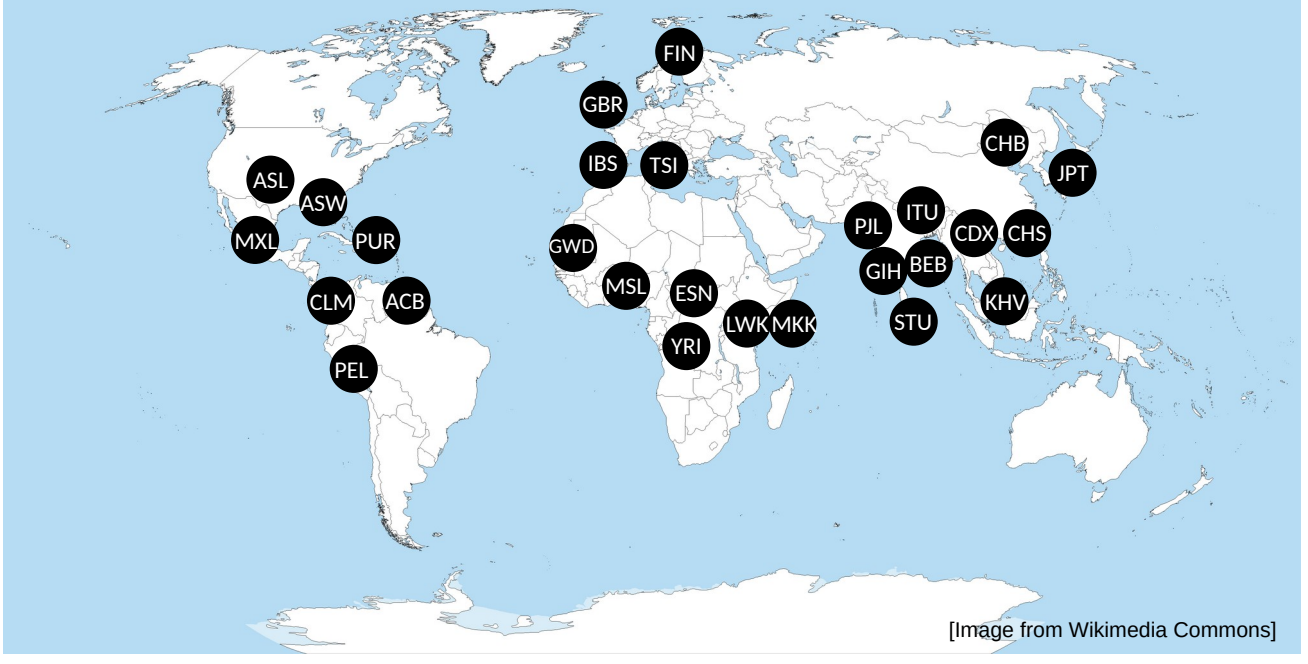
Human pangenome

- Liao et al 2023: 94 haplotypes
- Lucas et al 2026: 460 haplotypes



Human pangenome

- Liao et al 2023: 94 haplotypes
- Lucas et al 2026: 460 haplotypes

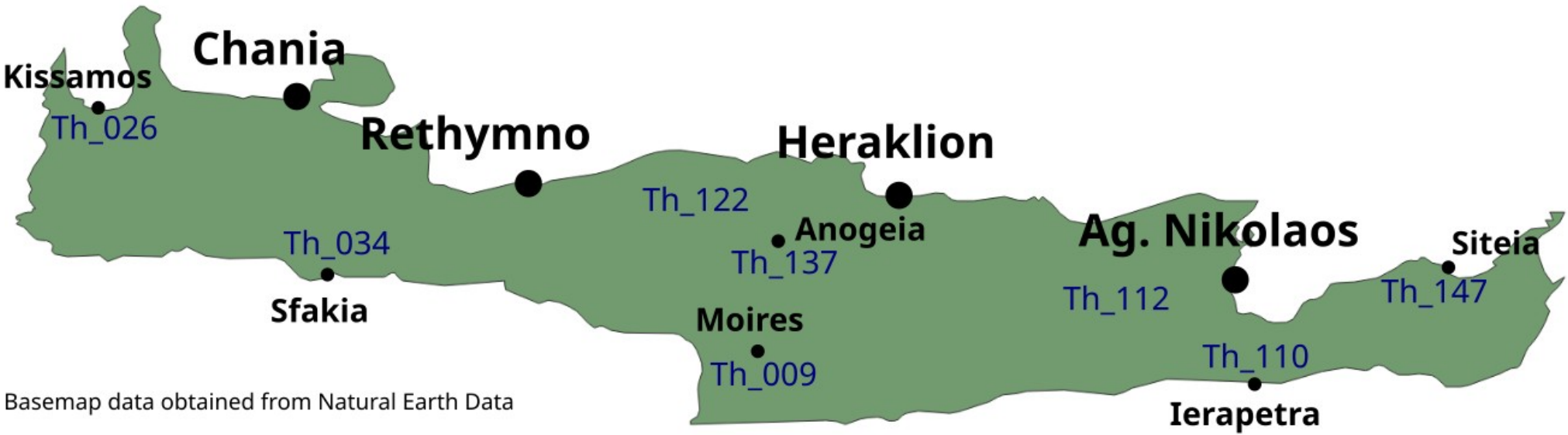


Genomes of Greek populations

- Whole genomes based on **short reads**
  - HELIC-MANOLIS and HELIC-Pomak (Gilly et al 2018)
  - Genome of Greece (Patrinos et al 2020)
  - Simons Genome Diversity Project (Mallick et al 2016)
- Study only SNPs and small indels
- **However, no long read Greek genomes…**

The THESEUS project

- Sequencing the Cretan genome
- 8 individuals with HiFi long reads (~50x)



Analysis workflow

***De novo* assembly**

Genome assembly

- Hifiasm
- Flye
- Shasta

Assembly evaluation

- BUSCO
- MerquryFK
- Blobtools

Analysis workflow

***De novo* assembly** → **Structural variant (SV) calling**

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- Genomic variants
- SNPs
- Small indels
- Large indels = Structural Variants

- Large means >50 nucleotides

- Majority of genetic diversity
- More likely to affect a gene

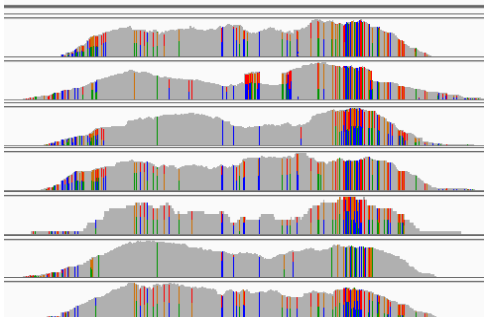
Analysis workflow

***De novo* assembly** → **Structural variant (SV) calling**

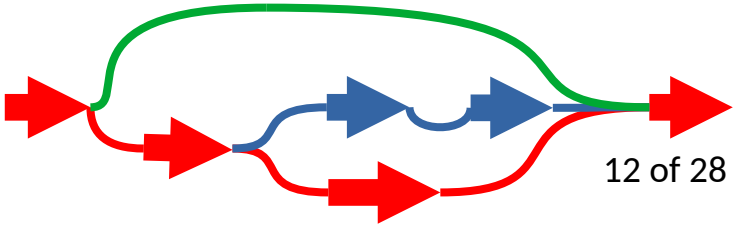
Genome assembly
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Single reference genome
- ensemble SV calling
  - 6 read-based methods
  - 6 assembly-based methods



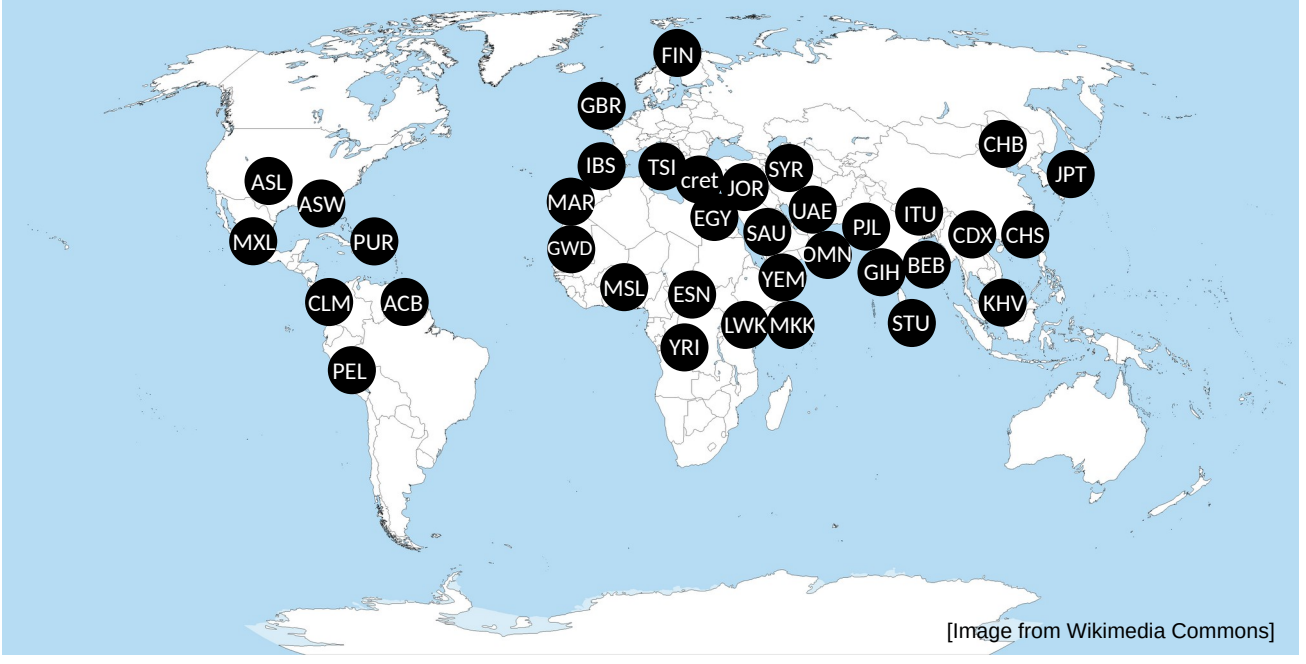
Pangenome graph
- pre-built HPRC2 graph
  - 464 haplotypes
- add 8 x 3 Cretans
- add 106 Arabs



Comparison of Cretans with other populations

Cretans in the world

- HPRC2 + Cretans + Arabs =
  594 haplotypes, 36 populations



Cretans in the world

- **159,400 total** SVs in the pangenome graph
- For **60,047** SVs
  - confident allele calls for ALL 594 haplotypes
- Cluster the 594 haplotypes based on their **SV profile**

Cretans in the world

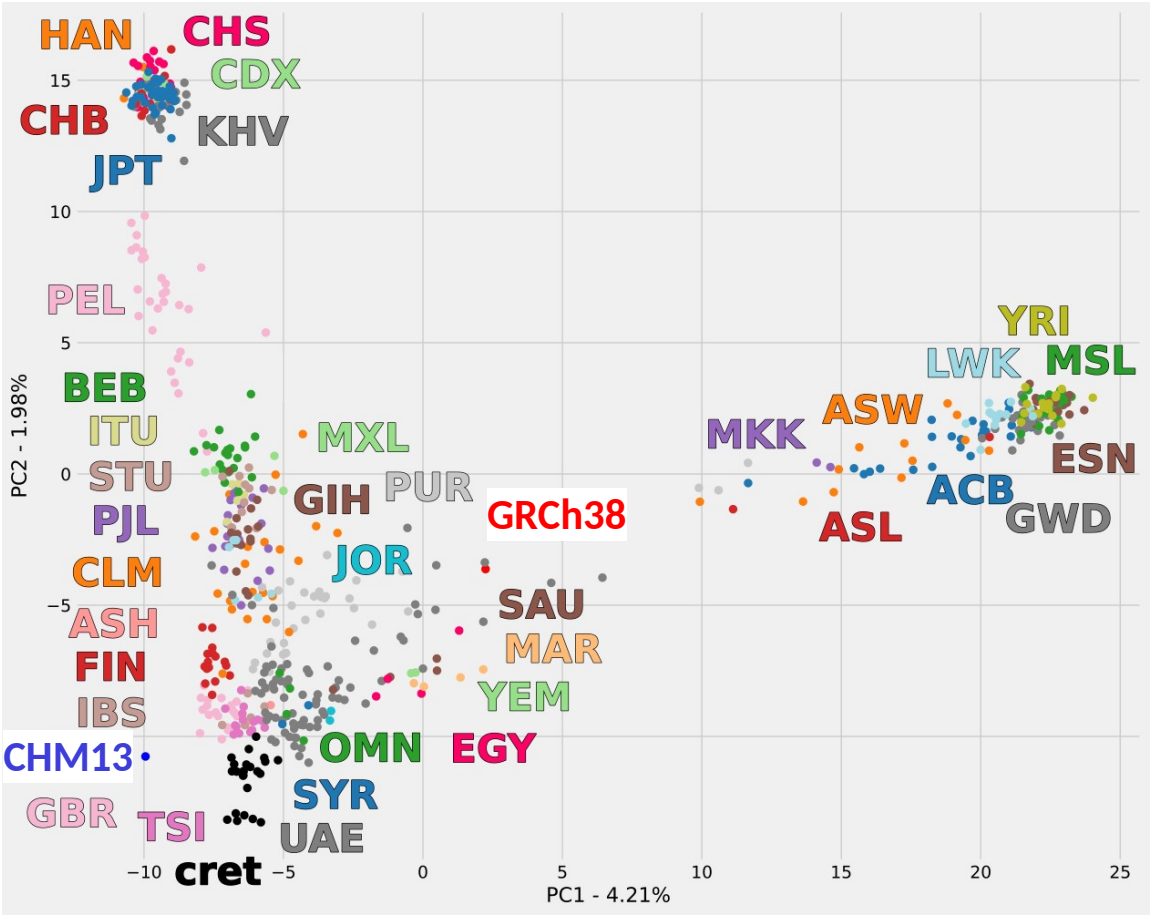
	hap_1	hap_2	hap_3	hap_4	hap_5
SV_1	0	0	1	2	1
SV_2	0	1	2	0	0
SV_3	.	2	0	1	2
SV_4	1	1	0	0	0
SV_5	1	1	.	.	0
SV_6	0	0	2	1	2

missing data

- exclude SV_3 and SV_5

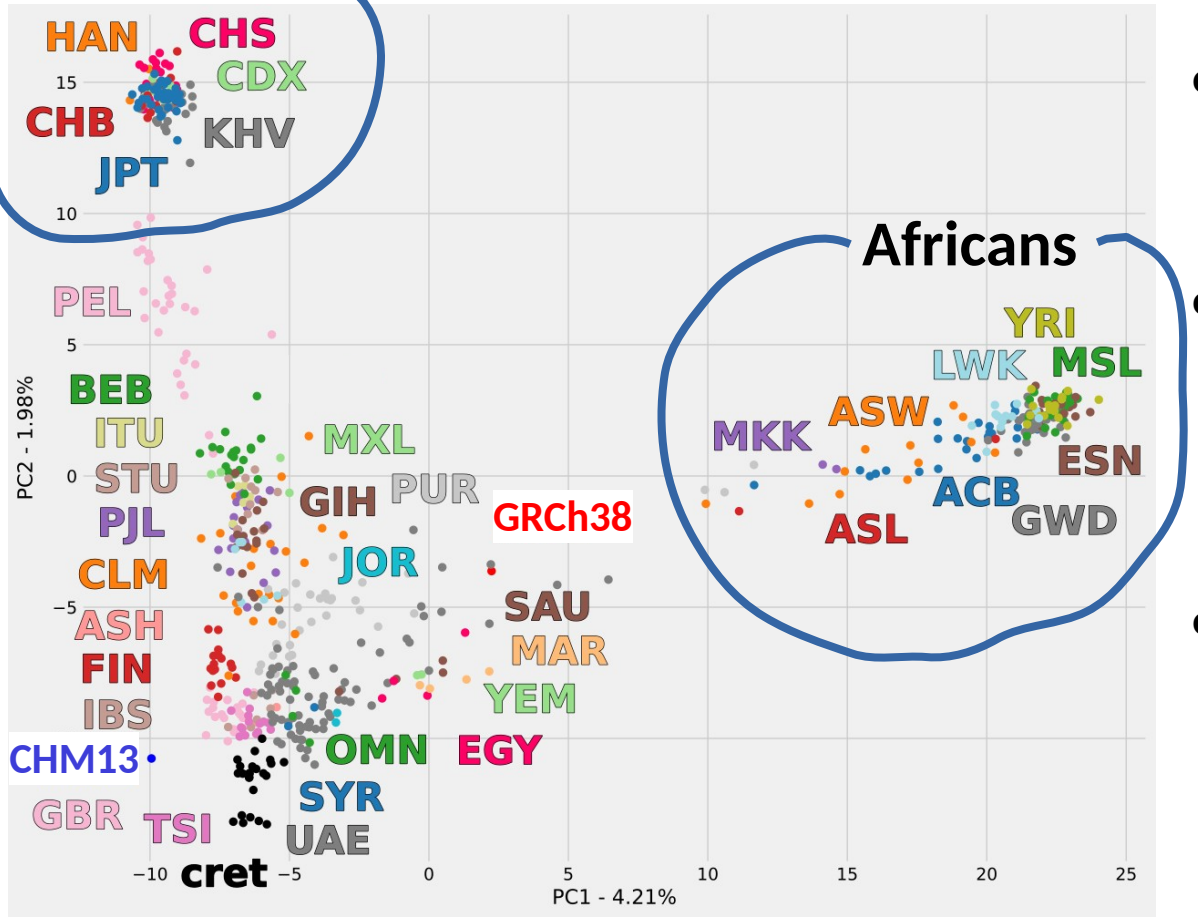
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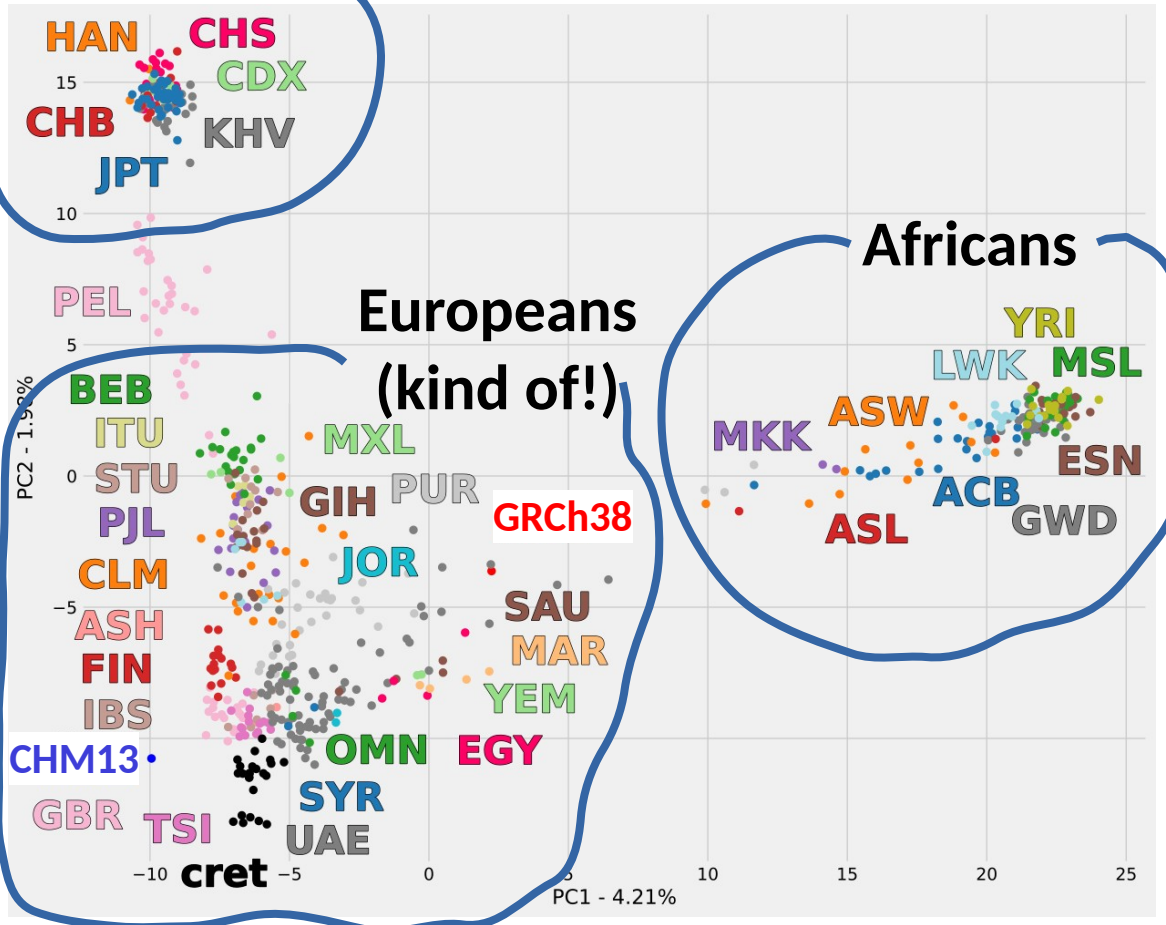
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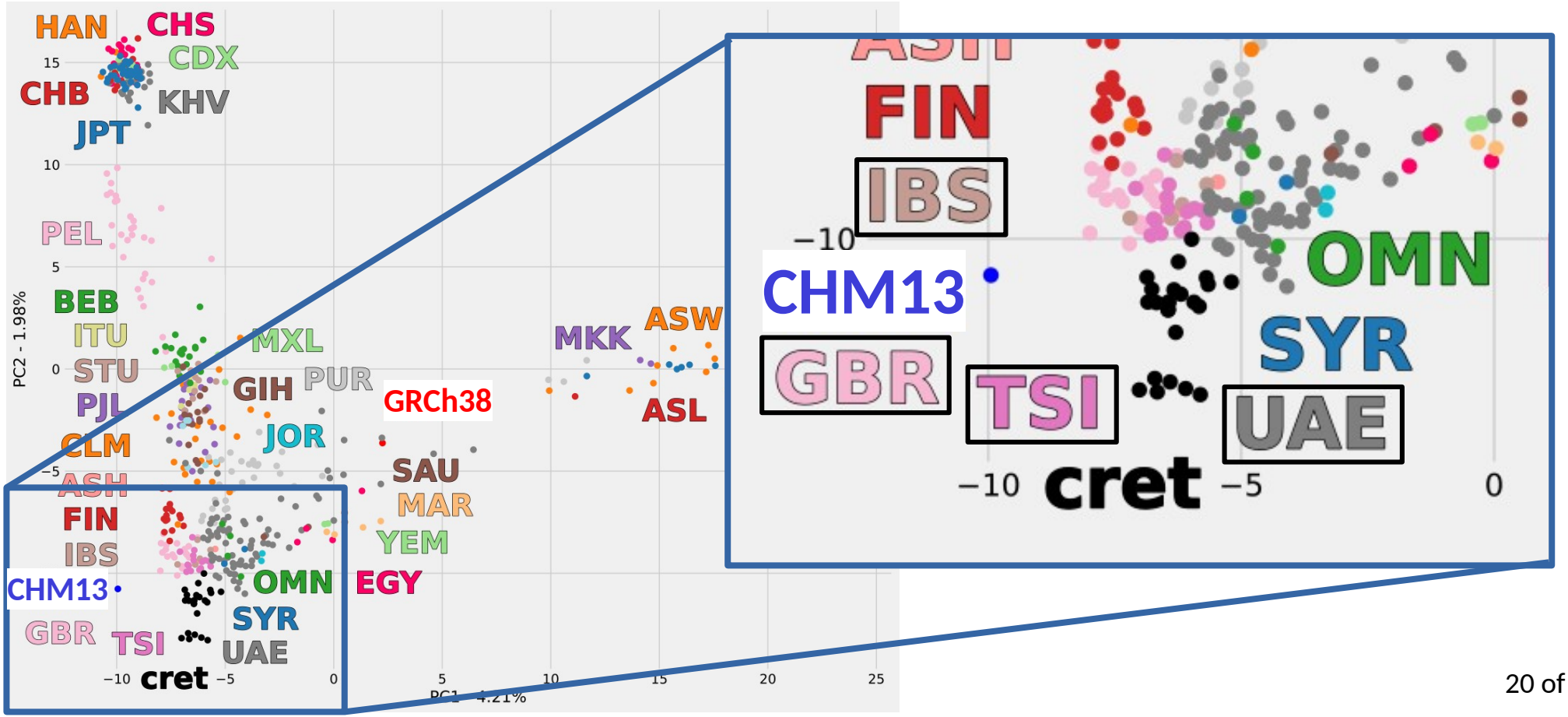
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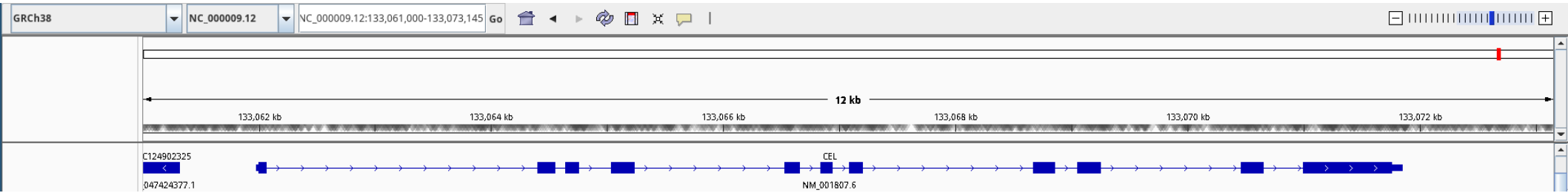
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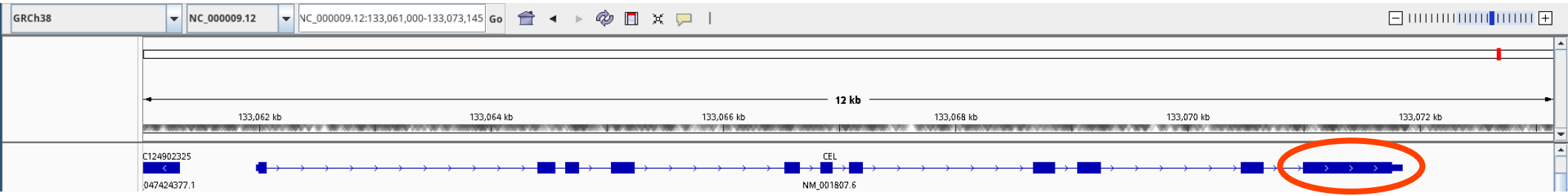
The case of the Carboxyl Ester Lipase

Carboxyl Ester Lipase



- chromosome 9: 133,062,000-133,072,000
- catabolises lipids in the intestine

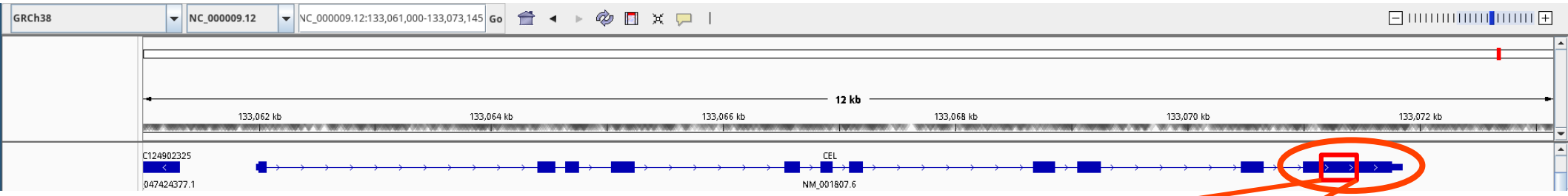
Carboxyl Ester Lipase



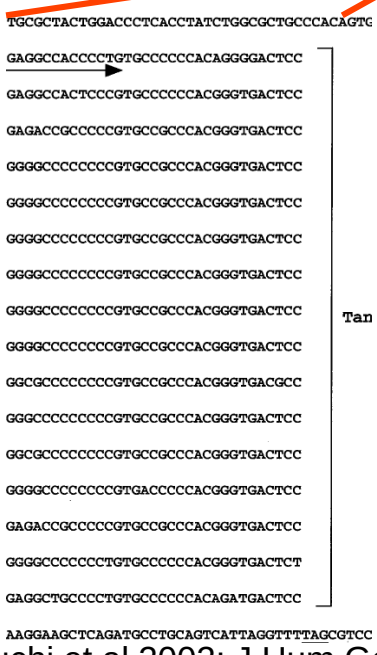
exon 11

- chromosome 9: 133,062,000-133,072,000
- catabolises lipids in the intestine
- Variable number tandem repeat in exon 11

Carboxyl Ester Lipase



- Variable number tandem repeat
  - Possible alleles
    - 16 repeats = C
    - 10-19 repeats
  - What are the Cretan alleles?



exon 11

16 repeats
||
Common allele

[Higuchi et al 2002; J Hum Genet]

Carboxyl Ester Lipase

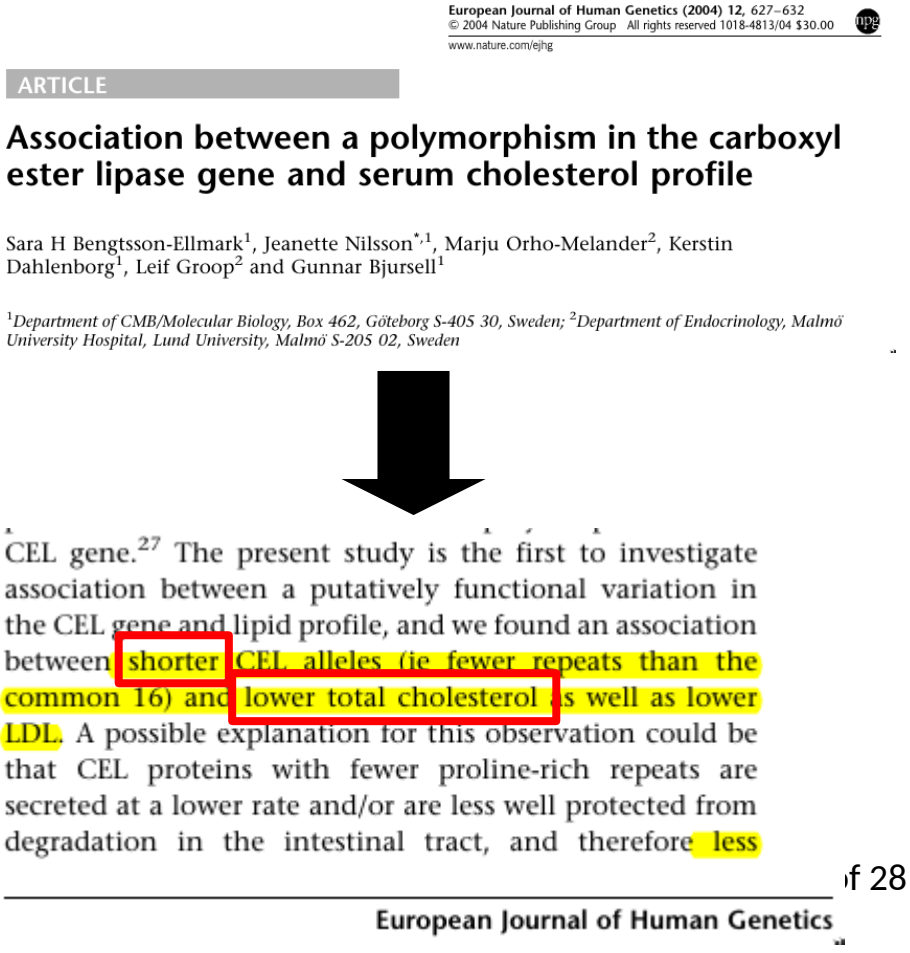
Sample	Region	Genotype*
Th_009	Moires	C-1/C-3
Th_026	Kissamos	C/C-3
Th_034	Sfakia	C-1/C-1
Th_110	Ierapetra	C/C-3?
Th_112	Lasithi	C-1/C-2
Th_122	Mylopotamos	C-2/C-2
Th_137	Anogeia	C/C-2
Th_147	Siteia	C-1/C-4

* C = common allele = 16 repeats

Carboxyl Ester Lipase

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ARTICLE

Association between a polymorphism in the carboxyl ester lipase gene and serum cholesterol profile

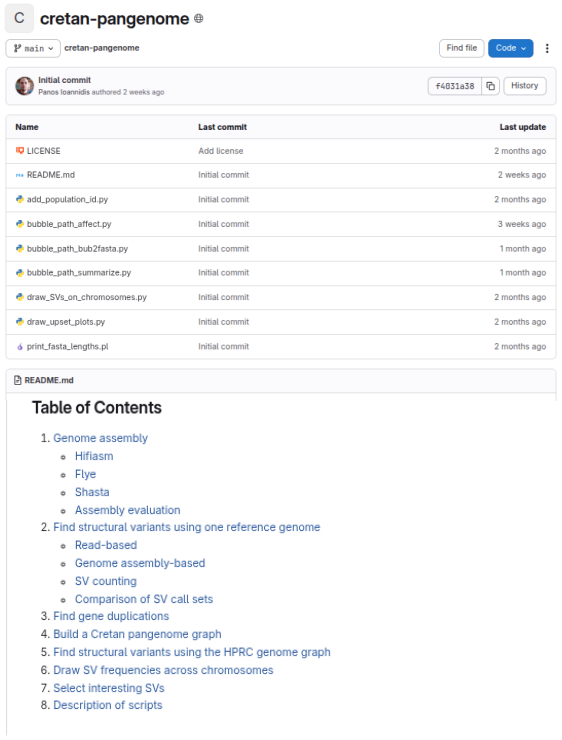
Sara H Bengtsson-Ellmark[1], Jeanette Nilsson*[,1], Marju Orho-Melander[2], Kerstin Dahlenborg[1], Leif Groop[2] and Gunnar Bjursell[1]

[1]*Department of CMB/Molecular Biology, Box 462, Göteborg S-405 30, Sweden;* [2]*Department of Endocrinology, Malmö University Hospital, Lund University, Malmö S-205 02, Sweden*

CEL gene.[27] The present study is the first to investigate association between a putatively functional variation in the CEL gene and lipid profile, and we found an association between shorter CEL alleles (ie fewer repeats than the common 16) and lower total cholesterol as well as lower LDL. A possible explanation for this observation could be that CEL proteins with fewer proline-rich repeats are secreted at a lower rate and/or are less well protected from degradation in the intestinal tract, and therefore less

Next steps?

- More interesting cases?
- Writing the manuscript
- Code/Methods available at gitlab



C cretan-pangenome

main · cretan-pangenome · Find file · Code

Initial commit
Panos Ioannidis authored 2 weeks ago · f4031a38 · History

Name	Last commit	Last update
LICENSE	Add license	2 months ago
README.md	Initial commit	2 weeks ago
add_population_id.py	Initial commit	2 months ago
bubble_path_affect.py	Initial commit	3 weeks ago
bubble_path_bub2fasta.py	Initial commit	1 month ago
bubble_path_summarize.py	Initial commit	1 month ago
draw_SVs_on_chromosomes.py	Initial commit	2 months ago
draw_upset_plots.py	Initial commit	2 months ago
print_fasta_lengths.pl	Initial commit	2 months ago

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   - Comparison of SV call sets
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7. Select interesting SVs
8. Description of scripts

https://gitlab.com/pioannidis/cretan-pangenome

Thank you - questions welcome!

